TECHNOLOGY OVERVIEW

Assembly Analysis Without the Preparation

How AsmFEA turns a rigged CAD assembly into a solved, checked and reported structural analysis

Abstract

Structural analysis of a single part is now routine. Analysis of a whole assembly is not. Before an assembly can be solved, someone has to decide how every part will be meshed, how each pair of parts is joined, which parts can move, and how big a model the computer can hold. Those decisions take longer than the solve, they need an experienced analyst, and most of them have to be made again every time the design changes.

AsmFEA is built to remove that preparation. It reads the assembly as the designer built it, with its mates, weldments, sheet metal, structural members and fasteners, and makes the analyst's decisions itself: how each part should be meshed, what each contact between parts means, whether the model is properly held, and how the run should be organised for the machine it is on. It solves with CalculiX, or for very large assemblies with its own part-by-part solver, then checks its own answer, grades its accuracy and writes the report. This paper describes what AsmFEA does and the decision path it follows for every model.

The Preparation Problem

An assembly is not one shape. It is a frame of welded tubes, plates bolted to brackets, a hinge pin in a bore, a sheet metal cover and a few machined blocks. Each of these wants different treatment. A long tube is best represented by elements laid out along its length. A thin plate needs elements through its thickness that can bend properly. A chunky casting needs a volume mesh. Using one approach everywhere either wastes most of the computer's effort or gives poor answers in the thin parts.

The joints between parts are harder still. Two faces that touch might be welded, bolted, free to slide, or simply resting on each other. A pin in a hole might be a hinge, or it might be locked by other mates. A distance mate positions a part but does not glue it. Getting these wrong does not cause an error message. It gives a model that solves and produces plausible but wrong numbers.

Then there is the computer. A detailed assembly can easily need more memory than the machine has. The usual answer is to simplify the model by hand until it fits, which brings back the judgement calls the analysis was supposed to settle.

Finally, the design changes. A tube gets thicker, a bracket moves, a gusset is added. In a hand-built analysis model much of the meshing, joint definition and load application has to be checked or rebuilt. The cost of keeping the analysis up to date is often why it falls behind the design.

The AsmFEA Approach

AsmFEA treats the assembly itself as the analysis model and moves every routine judgement into software, where it is made the same way every time and recorded in the report. Its main attributes are:

- **It works on the rigged assembly.** Mates, weldment members, sheet metal, fastener data and materials are read from the CAD model. No separate analysis geometry is built and no re-rigging is needed after a design change.
- **Connections are found and classified automatically.** Touching faces, mates, welds and fasteners are turned into bonded, hinged, sliding or bolted joints by fixed rules. The user can override any of them.
- **Each part is meshed in the way that suits it.** Structural members, thin walls, plates and solid parts each get their own route. If a specialised route cannot be used, the part falls back to a general volume mesh and the report says so.
- **The run is planned for the machine.** The user chooses whether to optimise for speed, balance or accuracy. AsmFEA looks at the model size, the free memory and the processor, and decides the mesh, the solver and the level of detail at the joints.
- **Large assemblies are solved part by part.** When a model will not fit in memory as one system, a dedicated solver condenses each part onto its joints and solves only the joints together. Memory is set by the largest part, not the whole assembly.
- **Every answer is checked and graded.** Each run is normally solved twice at two mesh densities, compared, and given an accuracy grade. Reactions are balanced against loads, and stress peaks at sharp corners and joint edges are recognised for what they are.
- **It fails loudly.** A missing material, an unmatched face, a floating part or a failed mesh is never filled in with a guess. It produces a plain-language explanation of what to change in the model.

The Benefits

The first benefit is time. The steps that usually dominate an assembly analysis, preparing geometry, meshing part by part and defining every joint, are done automatically in seconds to minutes. Re-running after a design change repeats only the stages whose inputs actually changed. A run in which nothing has changed returns in a few seconds, and a change to a pass/fail limit updates the results and the report without solving again.

The second benefit is consistency. Because connections, mesh routes and run settings are decided by rules, two people analysing the same assembly get the same model, and the report states every assumption that affects the result.

The third benefit is reach. Assemblies that would normally be cut down to a few parts can be analysed whole. On a desktop workstation with 64 GB of memory, AsmFEA has solved a welded ladder frame of 1.5 million nodes end to end in about six minutes, and a sectional water tank of thirteen million unknowns that would not fit in memory as a single system.

From CAD Assembly to Neutral Model

The CAD add-in does no analysis. Its only job is to describe the assembly faithfully in a neutral form: a model file plus one geometry file per unique part. The model file records each part's position, material, mates, the faces that loads and fixtures act on, and what kind of body each part is.

- A weldment member becomes a **member**, with its profile, wall thickness, path and the clean spans between its joints.
- A sheet metal body becomes **sheet**, with its thickness.

- A library fastener becomes a **fastener**, with its type, diameter and axis.
- Everything else is a **solid**. A member or fastener whose data cannot be read is treated as a solid rather than guessed at.

Faces are identified by a fingerprint of their type, area, centre and direction, so the analysis can find them again in the geometry files and after a design change. If a face cannot be matched, or two faces match equally well, the run stops and says which. A missing material is always a blocking problem: AsmFEA never substitutes a default.

The Decision Path

Every model goes through the same sequence of decisions. Each step either settles something about the model or stops with an explanation of what needs to change.

1 Match the geometry to the description

Each body in the model file is matched to its geometry by volume, centre and shape, and each referenced face by its fingerprint. A body whose volume differs by more than 1 % gives a warning. An unmatched or ambiguous body or face stops the run.

2 Find where parts meet

Pairs of parts whose bounding boxes overlap are searched face by face for flat faces that face each other and overlap, pins sitting in coaxial holes, and lines where a flat face meets a curved one. Two faces count as touching if they are within the bond tolerance, 0.5 mm by default, either as a gap or an overlap. Overlapping volumes above 1 mm^3 are reported as interference.

3 Decide what each contact means

The mates between two parts decide how their touching faces are treated. A user override always wins.

- Touching faces with no mate between the parts are **bonded**, as are bodies of one weldment and parts held by a lock mate.
- Distance and angle mates only position parts, so their touching faces still bond. Limit mates leave the parts free.
- A concentric or hinge mate on cylindrical faces becomes a **hinge**, but only if the other mates actually let the part turn. If they do not, the joint is bonded. With a hinge in place, the parts' other touching faces become **sliding** contacts that cannot separate.
- Slot, width and slider mates become sliding joints. Screw, gear and rack mates become couplings that link the motions of the two parts.
- A fastener passing through coaxial holes in two or more parts is a **bolted** joint. By default the clamped faces are tied. With a preload set, the bolt is modelled as a spring that carries the preload, and the clamped faces get frictional contact.
- Frictional contact is never chosen automatically. It is opted into connection by connection.
- Cosmetic weld beads bond the faces along the weld path. Welds on both sides or all round bond the whole interface; intermittent welds bond only the welded lengths.

Large rigs are rarely rigged with mates throughout, so for models of many parts AsmFEA works the other way round, as meshless tools do: it first **bonds everything by proximity** (overlapping parts included), then applies the mates as exceptions. A hinge or sliding joint is kept only where it frees a motion that matters: one between parts already joined by bonds, or on a pinned face, is bonded instead. This method is chosen automatically for models of more than 50 parts, or when the mate-driven rules leave a mechanism, and can be set in the study's advanced settings.

4 Check that the model is held

AsmFEA works out the free motions of the assembly from its fixtures and joints before any meshing. A group of parts connected to nothing and carrying no load is left out with a warning. A loaded part that is not held stops the run, because its answer would be meaningless. A part that is free to move as a mechanism is left out with its loads and a warning, unless the free motion involves a fixture or carries all of a load case, in which case the run stops and the parts involved are named.

On a model of many parts, small parts (under 0.1 % of the volume) that hang off the structure without carrying load through it are left out silently and listed in the report. Parts still floating after bonding are shown in a dialog with the gap to their nearest neighbour, where the bond tolerance can be raised or the parts left out of the analysis (they stay in SolidWorks).

5 Choose how to mesh each part

Each part is given one of three routes:

- **Member.** A structural member with clean spans is cut at its joints. The long spans are meshed by sweeping the cross-section along the path, and the end regions, where other parts attach, are swept through the wall. The pieces are tied together on the cut planes. The mesh is graded along each straight span: where a span runs clear of every joint (by one and a half section depths), its layers are four times longer and its cross-section coarser, while the windows round the joints keep the standard sizes. On a 174-part welded frame this made the model a third smaller with results unchanged to within the mesh's own variation, and cut the run from 69 to 42 minutes.
- **Skin.** Sheet metal, members without clean spans, and solids that are plate-like or bar-like are swept through their thickness. A solid qualifies when one dimension is at least four times the next and it fills at least 60 % of its bounding box. The two faces of the wall are paired, one face is meshed and the mesh is projected through the wall, giving elements that bend properly with one layer through the thickness.
- **Volume.** Every other part gets a quadratic tetrahedral mesh.

The element size follows the part's own size and the chosen fidelity. A swept route that cannot be used, because the wall thickness varies or the shape is not prismatic, falls back to a volume mesh. The run continues, the report says so, and the accuracy grade is lowered. A mesh with inverted or flat elements (a flat element is far stiffer than its neighbours and corrupts the part's answer) is refused and remade by another algorithm. A part that will not mesh is retried at smaller sizes before it is reported.

6 Plan the run for this machine

The user's choice of speed, balance or accuracy sets the mesh fidelity, whether a second, refined solve is made, whether parts that have not settled are solved a third time on a finer mesh, and how many

joints may be modelled exactly. The planner then measures the free memory, keeps a reserve for the rest of the computer (the larger of 3 GB and 4 % of the memory), and estimates what the model needs.

Optimise for	Mesh	Second, refined solve	Third, local solve	Joints modelled exactly
Speed	Coarse	No	No	None; all by smooth fields
Balanced	Standard	Yes	No	Smallest first, up to 10 000 joint unknowns per part and 60 000 in all
Accuracy	Fine	Yes	Where not converged	Up to 15 000 per part and 120 000 in all, when memory allows

If the whole model fits in memory as one system, it goes to CalculiX. If it does not, and the analysis is linear static, it goes to the part-by-part solver described below. Contact and buckling always go to CalculiX. The plan is shown before the run and can be overridden.

7 Build the solver model and solve

For CalculiX, independently meshed parts are joined by ties, with the finer side following the coarser. Hinges and sliding joints are built from rigid regions and constraint equations that leave exactly the intended motion free. Bolts with preload use a spring between the bearing faces plus a constant preload force. Each load case is a step. Where the loads allow it, each load is solved once and load cases are formed as weighted sums.

For linear buckling, each case has its own static preload step followed by the buckling step, so the load factor is measured against the actual loads.

When the planner asks for it, the whole model is solved twice: once at the base mesh and once with every element 0.6 times the size (except the long layers of graded member spans, which are already well within their accuracy and keep their size). The refined results are reported and the base results show how far the answer moved. With the third, local solve, the parts whose results moved by more than 10 % are meshed 0.6 times finer again and the model is solved once more; those three results also give, by Richardson extrapolation, the value the mesh is converging to and the band it is expected within.

8 Check the answer

Before results are shown, AsmFEA checks that the reactions balance the applied loads to within 1 %, that the displacements are not implausibly large, and that the solver did not quietly return the answer of a model that was not held. It compares each part between the mesh densities, on its displacement and on the stress it is assessed by (below), not on raw peaks, which never settle at a sharp corner: a part whose displacement settles but whose stress keeps rising is flagged as a stress singularity; a part whose results change by more than 10 % is flagged as poorly converged.

9 Assess stress the way an engineer would

Raw peak stresses at the edge of a bonded joint or at a sharp corner grow without limit as the mesh is refined, so judging a design on them is misleading. AsmFEA follows static design-code practice: only the sharp edge itself is set aside, and the stress all round it is assessed:

- At bonded joints, such as welds, the forces passed through the joint are turned back into a structural stress, split into membrane and bending parts over a short length of the joint. This is the established method for assessing welded joints.
- Everywhere, the most-stressed points are classified along lines through the wall into membrane and bending stress, following pressure-vessel code practice, round the joints as well as away from them. A local notch peak above yield therefore does not fail a static check on its own.
- At the edge of a joint on a thin wall, where the stress through the wall is linear and a line through it would return the edge's own spike, the surface stress is read as welded-joint practice reads it, between 0.4 and 1.6 wall thicknesses from the edge (widened to two elements where the elements at the joint are longer than that), and taken at 0.4 thicknesses for the static check. Extrapolated back to the edge it gives the hot-spot stress, a fatigue measure, which is kept for that purpose.
- The edge of a loaded face, where an evenly spread load stops abruptly, is set aside in the same way as a joint's edge: the stress there belongs to the idealised load, not the structure.
- Where the stress beside a joint keeps rising as the mesh is refined while the deflection has settled, it has no finite value in a linear model. If the part's peak away from the joint has settled, the factor of safety uses that instead, and the joint is listed for a weld or fatigue check.

The plots are scaled to the assessed stress; local peaks beyond it are shown in their own colour, so they do not hide everything else. The factor of safety is given on yield and, for information, on ultimate strength, the margin left before fracture. Factors of safety, deflection limits, buckling factors and convergence are each reported as pass, warn or fail, with an explanation of where the result governs and what to do about it.

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10 Report

The report puts the description, summary, results, conclusion and assumptions first; the supporting data (model, study settings, units, materials, loads and fixtures, connections, mesh, resultant forces, detailed results and input files) is a data report of its own, so the report stays short. Every assumption that affects the result, such as an automatically bonded contact or a part that fell back to a volume mesh, is stated in the report, and joints that need a weld or fatigue check are listed as recommendations. A one-page summary gives the figures without part names, for readers who do not know the model. Changing a pass/fail limit afterwards updates the checks, plots and report without solving again.

The Part-by-Part Solver

A detailed assembly can need far more memory as one system than a desktop computer has, even though each of its parts is modest. AsmFEA's hybrid solver uses that structure. It reads the same solver model that CalculiX would, and writes its results in the same format, so everything after the solve is unchanged.

- **Parts are condensed onto their joints.** Each part is assembled and factorised on its own, with its internal ties applied exactly. Its interior is then eliminated, leaving only how the part responds at the faces where it joins other parts.
- **Joints are exact where they are small.** A joint whose footprint has up to 3 000 nodes on its master side is joined exactly, node for node, as CalculiX ties it. Joints are taken smallest first, within the per-part and

total limits the planner has set.

- **Larger joints use smooth fields.** Both sides of a large joint follow one smooth displacement field over its footprint, described by a polynomial of fourth degree. This keeps the joint system small at the cost of some extra stiffness at the joint, which the accuracy grade accounts for.
- **Only the joints are solved together.** The condensed parts are assembled into one joint system. After it is solved, each part is recovered from its joint displacements, reusing its factorisation when memory allowed it to be kept.
- **Memory is managed, not exceeded.** Parts are solved several at a time within a measured memory budget, and wait their turn when memory is short. If another program takes memory during a run, the run slows down and a warning suggests closing programs, but it is never stopped. If the solver does run out of memory, it solves again with smooth joints only, one part at a time, and says so.

The hybrid solver handles linear static analysis with tied joints, hinges, sliding faces and pinned supports. A pin holds its face's mean motion, so the bore it sits in can deform, and its reaction is read directly from the solve. Models with contact, preloaded bolts or buckling are solved by CalculiX.

Fatigue

Repeated loading cracks steel and aluminium at stresses well below yield, and even a rig that only has to last a few hours can see thousands of cycles. AsmFEA assesses fatigue from the static results, without solving again, so a change to the loading history takes seconds.

- **The loading history is a spectrum.** Each row is a load case cycling between two percentages of itself, or between two load cases for a load that moves, a number of times per block; the block repeats over the design life. At every point the stress tensors at the two ends give the range, and damage is summed over the rows by Miner's rule. The fatigue factor is the factor on every stress at which the damage over the design life reaches 1.
- **Welds are read as the standards read them.** A bonded joint is assessed on the member side by the structural stress at the weld line (membrane plus bending from the joint's forces, on one design curve, BS 7608 class D by default), or by nominal stress with a detail category per joint. The wall the joint sits on is assessed with published stress concentration formulas for square hollow section T-joints within their range of validity, and otherwise as an indicative surface hot-spot reading from the model, marked as such.
- **Parent material uses S-N data for the material.** The SolidWorks material's own S-N curve is used when it has one; otherwise published measured data, or an estimate from the tensile strength with a design factor and a surface factor, marked as estimated. Points next to supports, ties and load edges are not read, a peak at a sharp inside corner is refused, and each governing peak is checked for convergence between the two meshes.
- **Preloaded bolts are assessed on their stress area,** with the preload applied by a pre-tension element, so the bolt sees only the share of the external load that its stiffness takes.

Fatigue has its own half-page summary, report and data report. They name the static results they were made from, and say so when those have changed since.

Knowing How Good the Answer Is

Every result carries an accuracy grade of High, Good or Indicative, worked out from how the model was actually solved rather than from a general claim:

- A converged run at standard or fine fidelity, with displacement changing by no more than 2 % and the assessed stress by no more than 5 % between the two meshes, is graded High.
- A coarse mesh, a run without the second solve, swept parts that fell back to volume meshes in thin walls, or joints represented by smooth fields limit the grade to Good.
- A quick preview run, a change of more than 10 % between the two meshes, or a model where most joints are smooth fields is graded Indicative.

Problems are reported as structured diagnostics, each with a code, its severity, the stage that found it, the parts and faces involved, the evidence, the likely causes and the actions to take. When a run cannot start or fails, AsmFEA lists the blocking problems in plain terms, each with the change that would let the model run and a button that highlights the parts concerned in the CAD model.

Re-running After a Design Change

Each stage of the pipeline records a fingerprint of everything it depends on: the relevant parts of the model, the geometry of each part, the settings, and the version of the software. On the next run a stage is skipped if its fingerprint is unchanged and its outputs are intact. Meshes are kept per part, so changing one bracket remeshes only that bracket, and identical parts share one mesh. Changing a load or a fixture re-solves without remeshing. In a measured case, a re-run with nothing changed took 2.3 seconds against 234 seconds for the first run, and a re-run after changing only a pass/fail limit took 17 seconds.

Verification

A model that cannot be trusted is worse than no model, so AsmFEA ships with a verification report that is re-run for each release from stored models.

CalculiX itself is checked first: all 101 of its official test decks reproduce their reference results, and the standard NAFEMS benchmarks LE1 and LE10 agree to within 0.4 %. AsmFEA's own pipeline is then checked against closed-form answers.

Case	Quantity	Difference from hand answer
Square hollow section cantilever	Tip deflection	+0.39 %
Square hollow section cantilever	Root stress	0.00 %
Bar under its own weight	Deflection	−0.04 %
Two tied parts as a cantilever	Tip deflection	+0.62 %
Pinned column	Buckling load	−0.41 %
Swept structural member	Deflection against beam theory	within 0.20 %

The part-by-part solver is checked against CalculiX on the same models. With small joints modelled exactly, the peak displacement of a welded frame with 96 welded joints agrees to within 0.05 %, a bolted bracket, a

weldment edge joint and a portal frame agree exactly to the printed figures, and a weldment tee agrees to within 0.12 %.

> Hinges, preloaded bolts, gears, racks and screws have each been prototyped on a two-part model with a hand answer before use, and checked that the intended force or motion is actually transmitted, not only that the solver accepts the input. A preloaded bolt, for example, carried 12 139 N against a hand estimate of 11 667 to 12 273 N.

Conclusion

For analysis to keep pace with design, it has to work on the design as it is, and it has to be cheap to repeat. AsmFEA does this by making the analyst's routine decisions in software: what each part is, how it should be meshed, what each contact between parts means, whether the model is held, and how to organise the run for the computer it is on. It then checks its own answer, says how good it is, and explains plainly when the model needs to change. The result is a tool that is:

- fast enough to run on every design change, because only what changed is redone, and
- simple enough to use occasionally, because the decisions that normally need a specialist are made consistently and written into the report.

AsmFEA technology overview, version 0.6.0. Internal use. Solver: CalculiX 2.22. Figures quoted are from the AsmFEA verification report and from measured runs on a desktop workstation with 8 cores and 64 GB of memory.